NATIONAL UNIVERSITY OF SINGAPORE
Department of Mathematics

MA1522 Linear Algebra for Computing **Tutorial 1**

1. (a) Find a linear equation in the variables x and y that has a general solution $x = 1+2t$ and $y = t$ where t is an arbitrary parameter.

   (b) Show that $x = t$ and $y = \frac{1}{2}t - \frac{1}{2}$, where t is an arbitrary parameter, is also a general solution for the equation constructed in part (a).

2. Find a linear equation in the variables x, y, and z that has a general solution

$$\begin{cases} x &= 3 - 4s + t \\ y &= s \\ z &= t \end{cases} \qquad s, t \in \mathbb{R} \ .$$

3. Solve the following linear systems.

   (a)
$$\begin{cases} 3x_1 + 2x_2 - 4x_3 = 3 \\ 2x_1 + 3x_2 + 3x_3 = 15 \\ 5x_1 - 3x_2 + x_3 = 14 \end{cases}$$

   (b)
$$\begin{cases} a + b - c - 2d = 0 \\ 2a + b - c + d = -2 \\ -a + b - 3c + d = 4 \end{cases}$$

   (c)
$$\begin{cases} x - 4y + 2z = -2 \\ x + 2y - 2z = -3 \\ x - y = 4 \end{cases}$$

4. Determine the values of a and b so that the linear system

$$\begin{cases} ax + bz = 2 \\ ax + ay + 4z = 4 \\ ay + 2z = b \end{cases}$$

   (a) has no solution;

   (b) has only one solution;

   (c) has infinitely many solutions and a general solution has one arbitrary parameter;

   (d) has infinitely many solutions and a general solution has two arbitrary parameters.

5. (a) Does an inconsistent linear system with more unknowns than equations exist?

(b) Does a linear system which has only one solution, but more equations than unknowns, exist?

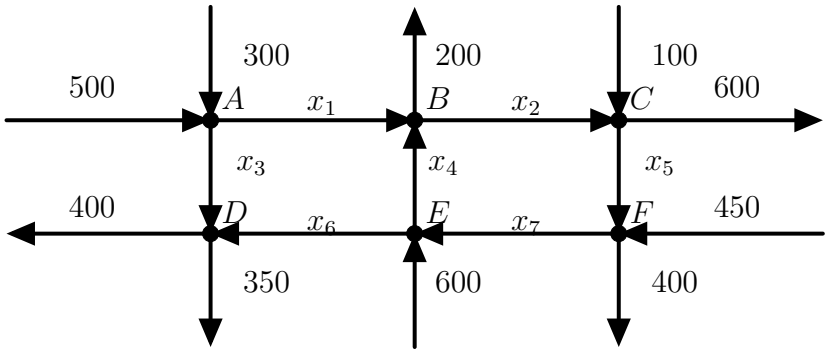
(c) Does a linear system which has only one solution, but more unknowns than equations, exists?

(d) Does a linear system which has infinitely many solutions, but more equations than unknowns, exists?

6. Solve the following system of non-linear equations:

$$\begin{array}{rcrcrcl} x^2 & - & y^2 & + & 2z^2 & = & 6 \\ 2x^2 & + & 2y^2 & - & 5z^2 & = & 3 \\ 2x^2 & + & 5y^2 & + & z^2 & = & 9 \end{array}$$

7. A network of one-way streets of a downtown section can be represented by the diagram below, with traffic flowing in the direction indicated. The average hourly volume of traffic entering and leaving this section during rush hour is given in the diagram.



(a) Do we have enough information to find the traffic volumes x_1, x_2, x_3, x_4, x_5, x_6, and x_7?

(b) Suppose $x_6 = 50$ and $x_7 = 100$. What is x_1, x_2, x_3, x_4, and x_5?

(c) Can the road between junction A and B be closed for construction while still keeping the traffic flowing in the same directions on the other streets? Explain.

Extra problems

1. The following is the reduced row-echelon form of the augmented matrix of a linear system:

$$\left(\begin{array}{ccc|c} a & b & c & d \\ 0 & e & f & g \\ 0 & 0 & h & k \end{array}\right),$$

where $a, b, c, d, e, f, g, h, k$ are constants. Suppose the solution set of this system is represented by a line that passes through the origin and the point $(1, 1, 1)$. Find the values of $a, b, c, d, e, f, g, h, k$. Justify your answer.

2. Determine which of the following statements are true. Justify your answer.

   (a) A homogeneous system can have a non-trivial solution.

   (b) A non-homogeneous system can have a trivial solution.

   (c) If a homogeneous system has the trivial solution, then it cannot have a non-trivial solution.

   (d) If a homogeneous system has a non-trivial solution, then it cannot have a trivial solution.

   (e) If a homogeneous system has a unique solution, then the solution has to be trivial.

   (f) If a homogeneous system has the trivial solution, then the solution has to be unique.

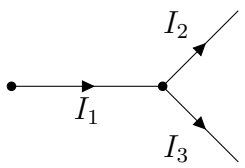
   (g) If a homogeneous system has a non-trivial solution, then there are infinitely many solutions to the system.

3. **(Application)** Electrical networks provides information about power sources, such as batteries, and devices powered by these sources, such as light bulbs or motors. A power source 'forces' a current of electrons to flow through the network, where it encounters various resistors, each of which requires that a certain amount of force be applied in order for the current to flow through it.

   The fundamental law of electricity is Ohm's law, which states exactly how much force E is needed to drive a current I through a resistor with resistance R. Ohm's law states $E = IR$, in other words, force = current $\times$ resistance. Here, force is measured in volts, resistance in ohms and current in amperes.

   The following two laws (discovery due to Kirchhoff), govern electrical networks. The first is a 'conservation of flow' law at each node; the second is a 'balancing of votage' law around each loop.

   (Kirchoff's Current Law (KCL)) At each node, the sum of the currents flowing into any node is equal to the sum of the currents flowing out of that node. For example, in the diagram below, by KCL, we have $I_1 = I_2 + I_3$.



(Kirchoff's Voltage Law (KVL)) In one traversal of any closed loop, the sum of the voltage rises equals to the sum of the voltage drops.

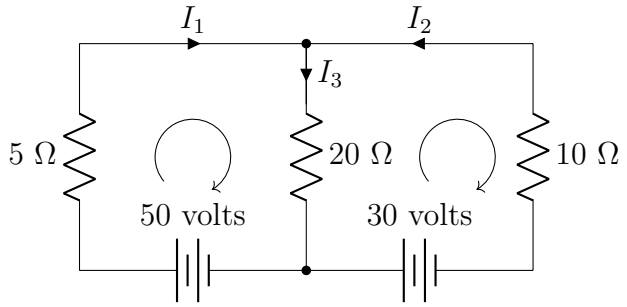
In circuits with multiple loops and batteries there is usually no way to tell in advance which way the currents are flowing, so the usual procedure in circuit analysis is to assign *arbitrary* directions to the current flows in the branches and let the mathematical computations determine whether the assignments are correct. In addition to assigning directions to the current flows, Kirchoff's Voltage Law requires a direction of travel for each closed loop. The choice is arbitrary, but for the sake of consistency we will always take this direction to be *clockwise*. We will also make the following conventions:

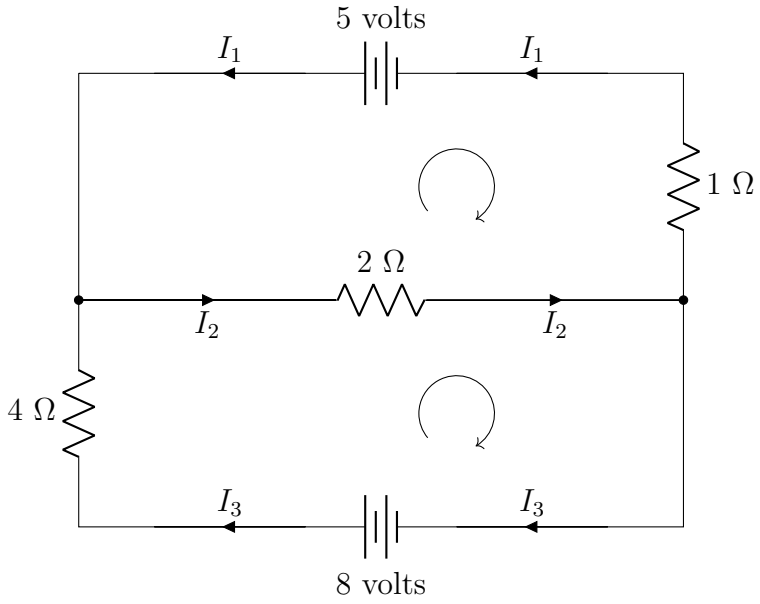
- A voltage drop occurs at a resistor if the direction assigned to the current through the resistor is the same as the direction assigned in the loop, and a voltage rise occurs at a resistor if the direction assigned to the current through the resistor is the opposite to that assigned in the loop.
- A voltage rise occurs at a battery if the direction assigned to the loop is from $-$ to $+$ through the battery, and a voltage drop occurs at a battery if the direction assigned to the loop is from $+$ to $-$ through the battery.

If we follow these conventions when calculating currents, then those currents whose directions were assigned correctly will have positive values and those whose direction were assigned incorrectly will have negative values.

For each of the following circuits, use KCL and KVL to write down a linear system with equations involving variables $I_1, I_2, \ldots$. Solve the linear system by Gaussian Elimination.

(a)





(b)

NATIONAL UNIVERSITY OF SINGAPORE
Department of Mathematics

MA1522 Linear Algebra for Computing **Tutorial 2**

1. Let $\mathbf{A}$ and $\mathbf{B}$ be $m \times n$ and $n \times p$ matrices respectively.

   (a) Suppose the homogeneous linear system $\mathbf{Bx} = \mathbf{0}$ has infinitely many solutions. How many solutions does the system $\mathbf{ABx} = \mathbf{0}$ have?

   (b) Suppose $\mathbf{Bx} = \mathbf{0}$ has only the trivial solution. Can we tell how many solutions are there for $\mathbf{ABx} = \mathbf{0}$.

2. (a) Let $\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$. Find a 4×3 matrix $\mathbf{X}$ such that $\mathbf{AX} = \mathbf{I}_3$.

   Hint: Write $\mathbf{X} = \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 & \mathbf{x}_3 \end{pmatrix}$, where $\mathbf{x}_i$ is a 4×1 matrix, for $i = 1, 2, 3$.

   (b) Let $\mathbf{B} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$. Find a 3×4 matrix $\mathbf{Y}$ such that $\mathbf{YB} = \mathbf{I}_3$.

3. (i) Reduce the following matrices $\mathbf{A}$ to its reduced row-echelon form $\mathbf{R}$.

   (ii) For each of the elementary row operation, write the corresponding elementary matrix.

   (iii) Write the matrices $\mathbf{A}$ in the form $\mathbf{E}_1\mathbf{E}_2 \dots \mathbf{E}_n\mathbf{R}$ where $\mathbf{E}_1, \mathbf{E_2}, \dots, \mathbf{E}_n$ are elementary matrices and $\mathbf{R}$ is the reduced row-echelon form of $\mathbf{A}$.

   (a) $\mathbf{A} = \begin{pmatrix} 5 & -2 & 6 & 0 \\ -2 & 1 & 3 & 1 \end{pmatrix}$.

   (b) $\mathbf{A} = \begin{pmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{pmatrix}$.

   (c) $\mathbf{A} = \begin{pmatrix} 1 & -1 & 0 \\ 2 & -2 & 1 \\ 1 & 2 & 3 \end{pmatrix}$.

4. Determine if the following matrices are invertible. If the matrix is invertible, find its inverse.

   (a) $\begin{pmatrix} -1 & 3 \\ 3 & -2 \end{pmatrix}$.

   (b) $\begin{pmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{pmatrix}$.

5. Write down the conditions so that the matrix $\begin{pmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{pmatrix}$ is invertible.

6. (a) Suppose $\mathbf{A}$ is a square matrix such that $\mathbf{A}^2 = \mathbf{0}$. Show that $\mathbf{I} - \mathbf{A}$ is invertible, with inverse $\mathbf{I} + \mathbf{A}$.

   (b) Suppose $\mathbf{A}^3 = \mathbf{0}$. Is $\mathbf{I} - \mathbf{A}$ invertible?

   (c) A square matrix $\mathbf{A}$ is said to be *nilpotent* if there is a positive integer n such that $\mathbf{A}^n = \mathbf{0}$. Show that if $\mathbf{A}$ is nilpotent, then $\mathbf{I} - \mathbf{A}$ is invertible.

Extra problems

1. Show that a linear system $\mathbf{Ax} = \mathbf{b}$ has either no solution, only one solution or infinitely many solutions. (Hint: Suppose $\mathbf{Ax} = \mathbf{b}$ has two different solutions $\mathbf{u}$ and $\mathbf{v}$. Use $\mathbf{u}$ and $\mathbf{v}$ to construct infinitely many solutions.)

2. Determine which of the following statements are true. Justify your answer.

   (a) If $\mathbf{A}$ and $\mathbf{B}$ are diagonal matrices of the same size, then $\mathbf{AB} = \mathbf{BA}$.

   (b) If $\mathbf{A}$ and $\mathbf{B}$ are square matrices of the same size, $(\mathbf{A} + \mathbf{B})^2 = \mathbf{A}^2 + \mathbf{B}^2 + 2\mathbf{AB}$.

   (c) If $\mathbf{A}$ is a square matrix, then $\frac{1}{2}(\mathbf{A} + \mathbf{A}^T)$ is symmetric.

   (d) If $\mathbf{A}$ and $\mathbf{B}$ are symmetric matrices of the same size, then $\mathbf{A} - \mathbf{B}$ is symmetric.

   (e) If $\mathbf{A}$ and $\mathbf{B}$ are symmetric matrices of the same size, then $\mathbf{AB}$ is symmetric.

   (f) If $\mathbf{A}$ is a square matrix such that $\mathbf{A}^2 = \mathbf{0}$, then $\mathbf{A} = \mathbf{0}$.

   (g) If $\mathbf{A}$ is an n by m matrix such that $\mathbf{AA}^T = \mathbf{0}$, then $\mathbf{A} = \mathbf{0}$.

3. Let $\mathbf{A}$ and $\mathbf{B}$ be two square matrices of the same order. Prove that if $\mathbf{A}$ is singular, then $\mathbf{AB}$ and $\mathbf{BA}$ are singular. (Prove the statement without using determinant.)

4. (**Polynomial Interpolation**)
   Given any n points in the xy-plane that has distinct x-coordinates, it is known that there is a unique polynomial of degree $n-1$ or less whose graph passes through those points. A degree $n-1$ polynomial has the following expression

$$y = a_0 + a_1 x + a_2 x^2 + \cdots + a_{n-1} x^{n-1}$$

   Suppose its graph passes through the points (x_1, y_1), (x_2, y_2),...,(x_n, y_n), it follows that the coordinates of the points must satisfy

$$\begin{array}{ccccccccc} a_0 & + & a_1 x_1 & + & a_2 x_1^2 & + \cdots + & a_{n-1} x_1^{n-1} & = & y_1 \\ a_0 & + & a_1 x_2 & + & a_2 x_2^2 & + \cdots + & a_{n-1} x_2^{n-1} & = & y_2 \\ \vdots & & \vdots & & \vdots & & \vdots & & \vdots \\ a_0 & + & a_1 x_n & + & a_2 x_n^2 & + \cdots + & a_{n-1} x_n^{n-1} & = & y_n \end{array}$$

   This is a linear system in the unknowns $a_0, a_1, ..., a_{n-1}$. The augmented matrix for the system is

$$\left(\begin{array}{ccccc|c} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} & y_1 \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} & y_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} & y_n \end{array}\right) \qquad \text{(V)}$$

   which has a unique solution whenever $x_1, x_2, ..., x_n$ are distinct.

   (a) Find a cubic polynomial whose graph passes through the points

x	1	2	3	4
y	3	-2	-5	0

(b) **(MATLAB)** The coefficient matrix of the linear system (V) is called a *Vandermonde Matrix*. The function fliplr(vander(v)) returns the Vandermonde matrix such that its rows are powers of the vector v. For example,

```
>> v=[1;2;3;4;5;6;7;8];

>> A=fliplr(vander(v))
```

will generate the following matrix

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 2^2 & 2^3 & 2^4 & 2^5 & 2^6 & 2^7 \\ 1 & 3 & 3^2 & 3^3 & 3^4 & 3^5 & 3^6 & 3^7 \\ 1 & 4 & 4^2 & 4^3 & 4^4 & 4^5 & 4^6 & 4^7 \\ 1 & 5 & 5^2 & 5^3 & 5^4 & 5^5 & 5^6 & 5^7 \\ 1 & 6 & 6^2 & 6^3 & 6^4 & 6^5 & 6^6 & 6^7 \\ 1 & 7 & 7^2 & 7^3 & 7^4 & 7^5 & 7^6 & 7^7 \\ 1 & 8 & 8^2 & 8^3 & 8^4 & 8^5 & 8^6 & 8^7 \end{pmatrix}$$

Use the Vandermonde matrix function to find a degree 7 polynomial that passes through

x	1	2	3	4	5	6	7	8
y	12	70	1244	10500	54268	205682	630540	1657024

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Department of Mathematics

MA1522 Linear Algebra for Computing **Tutorial 3**

1. Let $\mathbf{A}$ be the 4×4 matrix obtained from $\mathbf{I}$ by the following sequence of elementary row operations:

$$\mathbf{I} \xrightarrow{\frac{1}{2}R_2} \xrightarrow{R_1 - R_2} \xrightarrow{R_2 \leftrightarrow R_4} \xrightarrow{R_3 + 3R_1} \mathbf{A}.$$

   Write $\mathbf{A}^{-1}$ as a product of four elementary matrices.

2. Find an LU factorization for the matrices $\mathbf{A}$, and solve the equation $\mathbf{Ax} = \mathbf{b}$.

   (a) $\mathbf{A} = \begin{pmatrix} 2 & -1 & 2 \\ -6 & 0 & -2 \\ 8 & -1 & 5 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix}$.

   (b) $\mathbf{A} = \begin{pmatrix} 2 & -4 & 4 & -2 \\ 6 & -9 & 7 & -3 \\ -1 & -4 & 8 & 0 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 0 \\ 0 \\ 17 \end{pmatrix}$.

3. Let $\mathbf{A} = \begin{pmatrix} 2 & -6 & 6 \\ -4 & 5 & -7 \\ 3 & 5 & -1 \\ -6 & 4 & -8 \\ 8 & -3 & 9 \end{pmatrix}$.

   (a) Find an LU factorization of $\mathbf{A}$.

   (b) We can find a LU factorization in MATLAB using the command `lu`. Enter the following codes.

```
>> A=[2 -6 6;-4 5 -7;3 5 -1;-6 4 -8;8 -3 9];

>> [L U]=lu(sym(A)).
```

   Compare the results with the answer in (a).

4. Let $\mathbf{A} = \begin{pmatrix} -x & 1 & 0 \\ 0 & -x & 1 \\ 2 & -5 & 4-x \end{pmatrix}$. Compute the determinant of $\mathbf{A}$ and find all the values of x such that $\mathbf{A}$ is singular.

5. Show that $\begin{vmatrix} a+px & b+qx & c+rx \\ p+ux & q+vx & r+wx \\ u+ax & v+bx & w+cx \end{vmatrix} = (1+x^3)\begin{vmatrix} a & b & c \\ p & q & r \\ u & v & w \end{vmatrix}$.

6. Let $\mathbf{A} = \begin{pmatrix} 1 & 5 & 1 & 2 \\ 0 & 2 & 6 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 3 \end{pmatrix}$. Compute

   (a) $\det(3\mathbf{A}^T)$;

   (b) $\det(3\mathbf{A}\mathbf{B}^{-1})$; and

   (c) $\det((3\mathbf{B})^{-1})$.

7. Use Cramer's rule to solve

$$\begin{cases} x + 5y + 3z = 1 \\ \quad\; 2y - 2z = 2 \\ \quad\;\; y + 3z = 0 \end{cases}$$

8. Compute the adjoint of $\mathbf{A} = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 2 & 1 \\ 3 & 0 & 6 \end{pmatrix}$, and use it to compute $\mathbf{A}^{-1}$.

Extra problems

1. Let $\mathbf{A} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$.

   (a) Is it possible find an LU factorization of $\mathbf{A}$?

   (b) An $n \times n$ matrix $\mathbf{P}$ is a *permutation matrix* if there is exactly one 1 in each row and column, and 0 everywhere else, equivalent, a matrix obtained from the identity matrix using only row swap. Find a permutation matrix $\mathbf{P}$ such that $\mathbf{PA}$ has an LU factorization.

   This question demonstrates that not every matrix can be written as a LU factorization. However, we can always find a permutation matrix $\mathbf{P}$ such that $\mathbf{PA}$ has an LU factorization.

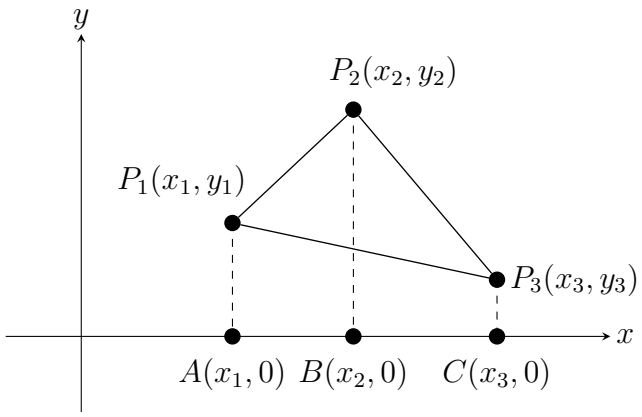
2. Let $\mathbf{A}$ be a $n \times n$ invertible matrix.

   (a) Show that $\text{adj}(\mathbf{A})$ is invertible and find its inverse.

   (b) Find $\det(\text{adj}(\mathbf{A}))$.

   (c) If $\det(\mathbf{A}) = 1$, show that $\text{adj}(\text{adj}(\mathbf{A})) = \mathbf{A}$.

3. Consider the triangle with vertices (x_1, y_1), (x_2, y_2) and (x_3, y_3) as shown in the figure below.



   We may compute the area of the triangle as

   (area of trapezoid AP_1P_2B)+(area of trapezoid BP_2P_3C)−(area of trapezoid AP_1P_3C)

   (a) Recall that the area of a trapezoid is $\frac{1}{2}$ the distance between the parallel sides of the trapezoid times the sum of the lengths of the parallel sides. Use this fact to show that the area of the triangle $P_1P_2P_3$ is

$$-\frac{1}{2}\left[(x_2y_3 - x_3y_2) - (x_1y_3 - x_3y_1) + (x_1y_2 - x_2y_1)\right].$$

(b) Show that the expression in the square brackets obtained in part (a) is the determinant of the following matrix

$$\mathbf{A} = \begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix}.$$

(c) Explain why we need to take the absolute value of $\det(\mathbf{A})$ before concluding that the area of the triangle is

$$\frac{1}{2}|\det(\mathbf{A})|.$$

(d) Find the area of the following quadrilaterals with the given vertices.

(i) P with vertices $(2,3)$, $(5,3)$, $(4,5)$, $(7,5)$.

(ii) Q with vertices $(-2,3)$, $(1,4)$, $(3,0)$, $(-1,-3)$.

4. (**Approximate integration**)

There are some integrals that are not possible to solve by finding the antiderivate. The integral can be approximated by using an interpolating polynomial to approximate the integrand and integrating the approximating polynomial. For example, suppose we want to evaluate the integral

$$\int_0^1 e^{-x^2} dx$$

We begin by picking a few points between the limits, say

$$x_0 = 0,\ x_1 = 0.25,\ x_2 = 0.5,\ x_3 = 0.75,\ x_4 = 1$$

and evaluate the integrand $f(x) = e^{-x^2}$ at these points. They are approximately

$$f(0) = 1, f(0.25) = 0.9394, f(0.5) = 0.7788, f(0.75) = 0.5698, f(1) = 0.3679$$

The interpolating polynomial is (check it!)

$$p(x) = 0.0416x^4 + 0.4882x^3 - 1.1846x^2 + 0.0226x + 1$$

and

$$\int_0^1 p(x)dx \approx 0.7468$$

which, up to the fourth decimal place, is exactly the integral $\int_0^1 e^{-x^2} dx$. The more points we pick, and thus the higher the degree of the polynomial, the more accurate the approximation becomes.

By using MATLAB, we can easily approximate the integral using interpolating polynomial. Suppose we want an interpolating polynomial of degree n. First create a vector whose entries are $n+1$ regular steps between a to b

```
>> v=[a:(b-a)/n:b]';
```

Then we create the Vandermonde matrix

```
>> A=fliplr(vander(v));
```

Let **b** be the vector whose entries are the evaluation of e^{-x^2} at the points of **v**

```
>> b=exp(-v.^2);
```

Then we obtain the coefficient of the interpolating polynomial of degree n

```
>> A\b
```

Use an interpolating polynomial of degree 10 to approximate the integral $\int_0^1 e^{-x^2} dx$.

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MA1522 Linear Algebra for Computing **Tutorial 4**

1. Let $A = \{\ (1+t, 1+2t, 1+3t) \mid t \in \mathbb{R}\ \}$ be a subset in $\mathbb{R}^3$.
   (a) Describe A geometrically.
   (b) Show that $A = \{\ (x, y, z) \mid x + y - z = 1 \text{ and } x - 2y + z = 0\ \}$.
   (c) Write down a matrix equation $\mathbf{Mx} = \mathbf{b}$ where $\mathbf{M}$ is a 3×3 matrix and $\mathbf{b}$ is a 3×1 matrix such that its solution set is A.

2. Let $\mathbf{u}_1 = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 3 \end{pmatrix}$, $\mathbf{u}_2 = \begin{pmatrix} 3 \\ -1 \\ 5 \\ 2 \end{pmatrix}$, and $\mathbf{u}_3 = \begin{pmatrix} -1 \\ 0 \\ 2 \\ 1 \end{pmatrix}$.
   (a) If possible, express each of the following vectors as a linear combination of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$.
   (i) $\begin{pmatrix} 2 \\ 3 \\ -7 \\ 3 \end{pmatrix}$ (ii) $\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$ (iii) $\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$ (iv) $\begin{pmatrix} -4 \\ 6 \\ -13 \\ 4 \end{pmatrix}$
   (b) Is it possible to find 2 vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ such that they are not a multiple of each other, and both are not a linear combination of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$?

3. Let $V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \,\middle|\, x - y - z = 0 \right\}$ be a subset of $\mathbb{R}^3$.
   (a) Let $S = \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 5 \\ 2 \\ 3 \end{pmatrix} \right\}$. Show that $\text{span}(S) = V$.
   (b) Let $T = S \cup \left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$. Show that $\text{span}(T) = \mathbb{R}^3$.

4. Which of the following sets S spans $\mathbb{R}^4$?
   (i) $S = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix} \right\}$.
   (ii) $S = \left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}$.

(iii) $S = \left\{ \begin{pmatrix} 6 \\ 4 \\ -2 \\ 4 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 2 \\ -1 \\ 2 \end{pmatrix}, \begin{pmatrix} 5 \\ 6 \\ -3 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 4 \\ -2 \\ -1 \end{pmatrix} \right\}$.

(iv) $S = \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} \right\}$.

5. Determine whether $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} \subseteq \text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$ and/or $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\} \subseteq \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ if

(a) $\mathbf{u}_1 = \begin{pmatrix} 2 \\ -2 \\ 0 \end{pmatrix}, \mathbf{u}_2 = \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}, \mathbf{u}_3 = \begin{pmatrix} 0 \\ 0 \\ 9 \end{pmatrix}, \mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \\ -5 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$.

(b) $\mathbf{u}_1 = \begin{pmatrix} 1 \\ 6 \\ 4 \end{pmatrix}, \mathbf{u}_2 = \begin{pmatrix} 2 \\ 4 \\ -1 \end{pmatrix}, \mathbf{u}_3 = \begin{pmatrix} -1 \\ 2 \\ 5 \end{pmatrix}, \mathbf{v}_1 = \begin{pmatrix} 1 \\ -2 \\ -5 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 0 \\ 8 \\ 9 \end{pmatrix}$.

6. Determine which of the following sets are subspaces. For those sets that are subspaces, express the set as a linear span. For those sets that are not, explain why.

(a) $S = \left\{ \begin{pmatrix} p \\ q \\ p \\ q \end{pmatrix} \middle| \, p, q \in \mathbb{R} \right\}$.

(b) $S = \left\{ \begin{pmatrix} a \\ b \\ c \end{pmatrix} \middle| \, a \geq b \text{ or } b \geq c \right\}$.

(c) $S = \left\{ \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} \middle| \, 4x = 3y \text{ and } 2x = -3w \right\}$.

(d) $S = \left\{ \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} \middle| \, \begin{vmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ a & b & c & d \end{vmatrix} = 0 \right\}$.

(e) $S = \left\{ \begin{pmatrix} w \\ x \\ y \\ z \end{pmatrix} \middle| \, w + x = y + z \right\}$.

(f) $S = \left\{ \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} \middle| \, ab = cd \right\}$.

(g) S is the solution set of $\mathbf{Ax} = \mathbf{0}$ where $\mathbf{A} = \begin{pmatrix} 2 & 2 & -1 & 0 & 1 \\ -1 & -1 & 2 & -3 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & -2 & 0 & -1 \end{pmatrix}$.

Extra problems

1. (a) Suppose $\mathbf{A} = \mathbf{PDP}^{-1}$ for some invertible matrix $\mathbf{P}$. Show that $\det(\mathbf{A}) = \det(\mathbf{D})$.

   (b) Suppose $\mathbf{A} = \mathbf{PDP}^{-1}$ for some invertible matrix $\mathbf{P}$ and $\mathbf{D}$ is a diagonal matrix. Show that $\mathbf{A}$ is invertible if and only if all the diagonal entries of $\mathbf{D}$ is nonzero.

   (c) Recall that a square matrix $\mathbf{A}$ is nilpotent if there is a positive integer k such that $\mathbf{A}^k = \mathbf{0}$. Show that if $\mathbf{A}$ is nilpotent, then $\det(\mathbf{A}) = 0$.

   (d) A square matrix is an *orthogonal* matrix if $\mathbf{A}^T = \mathbf{A}^{-1}$. Show that if $\mathbf{A}$ is orthogonal, then $\det(\mathbf{A}) = \pm 1$.

2. (a) Show that the solution set to any homogeneous linear system

   $$V = \{\ \mathbf{v} \in \mathbb{R}^n \mid \mathbf{Av} = \mathbf{0}\ \}$$

   is a subspace.

   (b) Let $V = \{\ \mathbf{v} \in \mathbb{R}^n \mid \mathbf{Av} = \mathbf{0}\ \}$. Show that if $\mathbf{Ax} = \mathbf{b}$ is consistent, then the solutions set is

   $$\mathbf{u}_p + V = \{\ \mathbf{u}_p + \mathbf{v} \mid \mathbf{v} \in V\ \},$$

   where $\mathbf{u}_p$ is a particular solution of $\mathbf{Ax} = \mathbf{b}$. (cf. Tutorial 1 Question 1)

   A subset of $\mathbb{R}^n$ is called an *affine space* if it is of the form $\{\ \mathbf{u} + \mathbf{v} \mid \mathbf{v} \in V\ \}$ for some subspace $V \subseteq \mathbb{R}^n$. Geometrically, an affine space is a subset of $\mathbb{R}^n$ that is parallel to a subspace. This exercise shows that the solution set to the linear system $\mathbf{Ax} = \mathbf{b}$ is an affine space $\{\ \mathbf{u}_p + \mathbf{v} \mid \mathbf{v} \in V\ \}$, where V is the solutions to homogeneous linear system $\mathbf{Ax} = \mathbf{0}$, and $\mathbf{u}_p$ is any particular solution.

3. Determine which of the following statements are true. Justify your answer.

   (a) If S_1 and S_2 are two subsets of $\mathbb{R}^n$, then $\operatorname{span}(S_1 \cap S_2) = \operatorname{span}(S_1) \cap \operatorname{span}(S_2)$.

   (b) If S_1 and S_2 are two subsets of $\mathbb{R}^n$, then $\operatorname{span}(S_1 \cup S_2) = \operatorname{span}(S_1) \cup \operatorname{span}(S_2)$.

4. In computers, information is stored and processed in the form of strings of binary digits, 0 and 1. For this exercise, we will work in the "world" of binary digits

   $$\mathbb{B} = \{0, 1\}.$$

   Addition in $\mathbb{B}$ works just as it does in $\mathbb{R}$, save for one special rule:

   $$1 + 1 = 0.$$

   We can similarly perform scalar multiplication in $\mathbb{B}$—however, note that in our "binary world", we only have two possible scalars: 0 and 1 (as opposed to any real number).

   Remark. The special rule for binary addition is equivalent to performing our standard operations **modulo 2**. That is, in our "binary world," we evaluate a sum according to its remainder when divided by 2: if the remainder is 0 (i.e., when a number is even), then it corresponds to the binary digit 0, and if the remainder is 1 (i.e., when a number is odd), then it corresponds to the binary digit 1.

1. Using the rules on the basic operations in $\mathbb{B}$, complete the addition and multiplication tables below.

$+$	0	1
0		
1		

$\times$	0	1
0		
1		

2. Recall that we created the Euclidean space $\mathbb{R}^n$ by taking the set of all n-vectors with real components (i.e., with components in $\mathbb{R}$). We can create the set $\mathbb{B}^n$ in a similar fashion, by taking the set of all n-vectors whose components are binary digits, 0 or 1. Observe, then, that the basic properties of addition and scalar multiplication in $\mathbb{R}^n$ directly apply to $\mathbb{B}^n$, as long as we remember that $1 + 1 = 0$ and the only scalars we are allowed to multiply by are 0 and 1.

   (a) Consider the Euclidean 3-space $\mathbb{R}^3$, which has infinitely many vectors. How many vectors does $\mathbb{B}^3$ have?

   (b) A *byte*—the fundamental unit of data used by many computers—is a string of 8 binary digits. Observe that we can treat each byte as a vector in $\mathbb{B}^8$. How many distinct bytes exist; that is, how many vectors are there in $\mathbb{B}^8$? How does this compare to Euclidean 8-space $\mathbb{R}^8$?

   (c) The Euclidean n-space $\mathbb{R}^n$ has infinitely many vectors. More generally, how many vectors are there in $\mathbb{B}^n$?

   For the purposes of this exercise, you may assume that $\mathbb{B}^n$ has all the properties of a subspace—that is, $\mathbb{B}^n$ is closed under addition and scalar multiplication. (Try to prove this yourself!)

3. To get a sense of how vectors work in $\mathbb{B}^n$, we take a simple example. Let's begin by working in $\mathbb{B}^3$—the set of all 3-vectors whose components are binary digits.

   (a) Let $S = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$ be the set of standard unit vectors in $\mathbb{R}^3$. Show that S forms a basis for $\mathbb{B}^3$.

NATIONAL UNIVERSITY OF SINGAPORE
Department of Mathematics

MA1522 Linear Algebra for Computing **Tutorial 5**

1. For each of the following sets of vectors S,

   (i) Determine if S is linearly independent.

   (ii) If S is linearly dependent, express one of the vectors in S as a linear combination of the others.

   (a) $S = \left\{ \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \\ 3 \end{pmatrix}, \begin{pmatrix} 3 \\ 6 \\ 6 \end{pmatrix} \right\}$.

   (b) $S = \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 4 \\ 2 \end{pmatrix} \right\}$.

   (c) $S = \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 4 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\}$.

   (d) $S = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \right\}$.

2. Suppose $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are linearly independent vectors in $\mathbb{R}^n$. Determine which of the sets S_1 to S_5 are linearly independent.

   (a) $S_1 = \{\mathbf{u}, \mathbf{v}\}$,

   (b) $S_2 = \{\mathbf{u} - \mathbf{v}, \mathbf{v} - \mathbf{w}, \mathbf{w} - \mathbf{u}\}$,

   (c) $S_3 = \{\mathbf{u} - \mathbf{v}, \mathbf{v} - \mathbf{w}, \mathbf{u} + \mathbf{w}\}$,

   (d) $S_4 = \{\mathbf{u}, \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} + \mathbf{w}\}$.

   (e) $S_5 = \{\mathbf{u} + \mathbf{v}, \mathbf{v} + \mathbf{w}, \mathbf{u} + \mathbf{w}, \mathbf{u} + \mathbf{v} + \mathbf{w}\}$.

3. For each of the following subspaces V, write down a basis for V.

   (a) $V = \left\{ \left. \begin{pmatrix} a+b \\ a+c \\ c+d \\ b+d \end{pmatrix} \right| a, b, c, d \in \mathbb{R} \right\}$.

   (b) $V = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right\}$.

(c) V is the solution space of the following homogeneous linear system

$$\begin{cases} a_1 & & + & a_3 & + & a_4 & - & a_5 & = & 0 \\ & a_2 & + & a_3 & + & 2a_4 & + & a_5 & = & 0 \\ a_1 & + a_2 & + & 2a_3 & + & a_4 & - & 2a_5 & = & 0 \end{cases}$$

4. For what values of a will $\mathbf{u}_1 = \begin{pmatrix} a \\ 1 \\ -1 \end{pmatrix}, \mathbf{u}_2 = \begin{pmatrix} -1 \\ a \\ 1 \end{pmatrix}, \mathbf{u}_3 = \begin{pmatrix} 1 \\ -1 \\ a \end{pmatrix}$ form a basis for $\mathbb{R}^3$?

5. Let U and V be subspaces of $\mathbb{R}^n$. We define the sum $U + V$ to be the set of vectors

$$\{\ \mathbf{u} + \mathbf{v} \mid \mathbf{u} \in U \text{ and } \mathbf{v} \in V\ \}.$$

Suppose $U = \text{span}\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 2 \\ 1 \end{pmatrix} \right\}$, $V = \text{span}\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 2 \\ -1 \end{pmatrix} \right\}$.

(a) Is $U \cup V$ a subspace of $\mathbb{R}^4$?

(b) Show that $U + V$ a subspace by showing that it can be written as a span of a set. What is the dimension?

(c) Show that $U+V$ contains U and V. This shows that $U+V$ is a subspace containing $U \cup V$.

(d) What are the dimensions of U and V?

(e) Show that $U \cap V$ a subspace by showing that it can be written as a span of a set. What is the dimension?

(f) Verify that $\dim(U + V) = \dim(U) + \dim(V) - \dim(U \cap V)$.

Extra problems

1. Let $\mathbf{u}_1, \mathbf{u}_2, ..., \mathbf{u}_k$ be vectors in $\mathbb{R}^n$ and $\mathbf{P}$ a square matrix of order n.

   (a) Show that if $\mathbf{Pu_1}, \mathbf{Pu_2}, ..., \mathbf{Pu_k}$ are linearly independent, then $\mathbf{u_1}, \mathbf{u_2}, ..., \mathbf{u_k}$ are linearly independent.

   (b) Suppose $\mathbf{u_1}, \mathbf{u_2}, ..., \mathbf{u_k}$ are linearly independent.

      (i) Show that if $\mathbf{P}$ is invertible, then $\mathbf{Pu_1}, \mathbf{Pu_2}, ..., \mathbf{Pu_k}$ are linearly independent.

      (ii) If $\mathbf{P}$ is not invertible, are $\mathbf{Pu_1}, \mathbf{Pu_2}, ..., \mathbf{Pu_k}$ are linearly independent?

2. Prove the theorem that $V \subseteq \mathbb{R}^n$ is a subspace if and only if $V = \text{span}\{\mathbf{u}_1, ..., \mathbf{u}_k\}$ for some $\mathbf{u}_i \in \mathbb{R}^n$, $i = 1, ..., k$.

3. (a) Let U and V be subspaces of $\mathbb{R}^n$. Prove that $U \cup V$ is a subspace if and only if $U \subseteq V$ or $V \subseteq U$.

   (b) Let U and V be subspaces of $\mathbb{R}^n$. Show that $\dim(U + V) = \dim(U) + \dim(V) - \dim(U \cap V)$.

4. **(MATLAB)** In the last tutorial, we introduced the "binary n-space" $\mathbb{B}^n$, which is governed by the special addition rule: $1+1 = 0$. Our goal in this problem is to investigate how working with binary vectors can help us detect and correct errors in information transmission.

   When a computer transmits a piece of information (a binary string) to another device, there is always a possibility of an error—that is, the receiving device might receive an incorrect binary string, perhaps due to external interference or noise in the communication channel. One way that a computer might "protect" its message is by adding extra information to the binary string so that the receiving device can detect—and ideally, correct—any errors that may have occured in transmission.

   Consider the following scenario: your friend Annette wants to send you a message $\boldsymbol{u}$. For the sake of simplicity, let's assume that Annette's message contains four bits—that is, $\boldsymbol{u}$ is a vector in $\mathbb{B}^4$ (as opposed to a standard *byte*, which is a vector in $\mathbb{B}^8$). Annette, however, is afraid that a transmission error might send you the wrong message—say, by accidentally changing a 1 to a 0, or a 0 to a 1.

   1. Rather than just sending you the message $\boldsymbol{u}$, Annette instead sends you the 8-vector that results when each bit in $\boldsymbol{u}$ is repeated twice. You receive the vector

      $$\left(\begin{array}{cc|cc|cc|cc} 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \end{array}\right),$$

      where divider bars have been used to split the string up into segments, each representing one bit in Annette's original message. Do you have enough information to decode Annette's original message?

      As the above situation suggests, a simple—perhaps naïve—error-correcting code would employ *repetition*: sending each bit repeatedly, with the hope that the recipient will be able to spot any errors. Observe, however, that this method significantly

increases the required amount of data to be transmitted. In the above example, Annette needed to send out twice as much data—this may be problematic for longer messages, specially since bandwidth is expensive!

2. Annette, who is running out of mobile data, attempts a different error-correcting code, invented by the 20th Century mathematician Richard Hamming. Recall that, taking the non-zero vectors in $\mathbb{B}^3$ as columns, we can create the *Hamming matrix*

$$\boldsymbol{H} = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 \end{pmatrix}.$$

From this, we formed the matrix $\boldsymbol{M}$ by taking the basis vectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3, \boldsymbol{v}_4$ for $S = \{\boldsymbol{x} \in \mathbb{B}^7 \mid \boldsymbol{H}\boldsymbol{x} = \boldsymbol{0}\}$ (i.e., the basis for the null space of $\boldsymbol{H}$) as its columns:

$$\boldsymbol{M} = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

(a) Explain why the set S is identical to the column space of $\boldsymbol{M}$. Hence, without explicitly calcuating any matrix products, explain why $\boldsymbol{H}(\boldsymbol{M}\boldsymbol{x}) = \boldsymbol{0}$ for all vectors $\boldsymbol{x} \in \mathbb{B}^4$.

(b) Consider the vector $\boldsymbol{v} = \boldsymbol{M}\boldsymbol{u}$, which is a vector in $\mathbb{B}^7$. The first three entries of $\boldsymbol{v}$ will later be used to detect errors. What are the last four entries of $\boldsymbol{v}$?

(c) Instead of transmitting the vector $\boldsymbol{u}$, Annette's computer instead sends out $\boldsymbol{v} = \boldsymbol{M}\boldsymbol{u}$. Assume that *at most* one error can occur during transmission.

- What vector will you receive if no errors occur during transmission?
- Let $\boldsymbol{e}_i$ denote the standard unit vector whose i-th entry is 1 and remaining entries are all 0's. Explain why you would receive a vector of the form $\boldsymbol{v} + \boldsymbol{e}_i$, for some $i \in \{1, \ldots, 7\}$, if an error has occured during transmission.

(d) Let $\boldsymbol{w}$ be the vector you receive on your device. Explain how calculating the matrix product $\boldsymbol{H}\boldsymbol{w}$ will allow you to detect and correct a potential transmission error. [*Hint:* First consider what the vector $\boldsymbol{w}$ would look like if no error has occured, then consider the possibility that an error has occured during transmission.]

(e) Your device receives the vector

$$\boldsymbol{w} = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

Has an error been made during transmission? Do you have enough information to deduce Annette's original message?

NATIONAL UNIVERSITY OF SINGAPORE
Department of Mathematics

MA1522 Linear Algebra for Computing **Tutorial 6**

1. (a) Let $\mathbf{u}_1 = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$, $\mathbf{u}_2 = \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}$, $\mathbf{u}_3 = \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix}$. Show that $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ forms a basis for $\mathbb{R}^3$.

   (b) Suppose $\mathbf{w} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$. Find the coordinate vector of $\mathbf{w}$ relative to S.

   (c) Let $T = \{\mathbf{v_1}, \mathbf{v_2}, \mathbf{v_3}\}$ be another basis for $\mathbb{R}^3$ where $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 5 \\ 4 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} -1 \\ 3 \\ 7 \end{pmatrix}$, $\mathbf{v}_3 = \begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix}$. Find the transition matrix from T to S.

   (d) Find the transition matrix from S to T.

   (e) Use the vector $\mathbf{w}$ in Part (b). Find the coordinate vector of $\mathbf{w}$ relative to T.

2. Let V be a subspace of $\mathbb{R}^n$ and $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ be a basis for a subspace V. Define $T = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ where

$$\mathbf{v}_1 = \mathbf{u}_1 + \mathbf{u}_2 + \mathbf{u}_3,\ \mathbf{v}_2 = \mathbf{u}_2 + \mathbf{u}_3 \text{ and } \mathbf{v}_3 = \mathbf{u}_2 - \mathbf{u}_3.$$

   (a) Show that T is a basis for V.

   (b) Find the transition matrix from S to T.

3. (a) Let $\mathbf{A} = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$. Is $\mathbf{b}$ in the column space of $\mathbf{A}$? If it is, express it as a linear combination of the columns of $\mathbf{A}$.

   (b) Let $\mathbf{A} = \begin{pmatrix} 1 & 9 & 1 \\ -1 & 3 & 1 \\ 1 & 1 & 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 5, 1, -1 \end{pmatrix}$. Is $\mathbf{b}$ in the row space of $\mathbf{A}$? If it is, express it as a linear combination of the rows of $\mathbf{A}$.

   (c) Let $\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 2 & 1 \\ 1 & 2 & 1 & 3 \\ 0 & 1 & 2 & 2 \end{pmatrix}$. Is the row space and column space of $\mathbf{A}$ the whole $\mathbb{R}^4$?

4. For each of the following matrices $\mathbf{A}$,

   (i) Find a basis for the row space of $\mathbf{A}$.

(ii) Find a basis for the column space of $\mathbf{A}$.

(iii) Find a basis for the nullspace of $\mathbf{A}$.

(iv) Hence determine rank($\mathbf{A}$), nullity($\mathbf{A}$) and verify the dimension theorem for matrices.

(v) Is $\mathbf{A}$ full rank?

(a) $\mathbf{A} = \begin{pmatrix} 1 & 2 & 5 & 3 \\ 1 & -4 & -1 & -9 \\ -1 & 0 & -3 & 1 \\ 2 & 1 & 7 & 0 \\ 0 & 1 & 1 & 2 \end{pmatrix}$

(b) $\mathbf{A} = \begin{pmatrix} 1 & 3 & 7 \\ 2 & 1 & 8 \\ 3 & -5 & -1 \\ 2 & -2 & 2 \\ 1 & 1 & 5 \end{pmatrix}$.

5. Let W be a subspace of $\mathbb{R}^5$ spanned by the following vectors

$$\mathbf{u}_1 = \begin{pmatrix} 1 \\ -2 \\ 0 \\ 0 \\ 3 \end{pmatrix}, \quad \mathbf{u}_2 \begin{pmatrix} 2 \\ -5 \\ -3 \\ -2 \\ 6 \end{pmatrix}, \quad \mathbf{u}_3 = \begin{pmatrix} 0 \\ 5 \\ 15 \\ 10 \\ 0 \end{pmatrix}, \quad \mathbf{u}_4 = \begin{pmatrix} 2 \\ 1 \\ 15 \\ 8 \\ 6 \end{pmatrix}.$$

(a) Find a basis for W.

(b) What is $\dim(W)$?

(c) Extend the basis W found in (a) to a basis for $\mathbb{R}^5$.

6. Let $S = \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 3 \\ 5 \\ 12 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 2 \\ 5 \end{pmatrix}, \begin{pmatrix} 3 \\ -1 \\ 1 \\ 4 \end{pmatrix} \right\}$ and $V = \text{span}(S)$. Find a subset $S' \subseteq S$ such that S' forms a basis for V.

Extra problems

1. Suppose $\mathbf{A}$ and $\mathbf{B}$ are two matrices such that $\mathbf{AB} = \mathbf{0}$. Show that the column space of $\mathbf{B}$ is contained in the nullspace of $\mathbf{A}$.

2. Let $\mathbf{A}$ be a $n \times m$ matrix and $\mathbf{P}$ an $n \times n$ matrix.

   (a) If $\mathbf{P}$ is invertible, show that $\text{rank}(\mathbf{PA}) = \text{rank}(\mathbf{A})$.

   (b) Given an example such that $\text{rank}(\mathbf{PA}) < \text{rank}(\mathbf{A})$.

   (c) If $\text{rank}(\mathbf{PA}) = \text{rank}(\mathbf{A})$. Can we conclude that $\mathbf{P}$ is invertible? Justify your answer.

3. Let $\mathbf{A} = \begin{pmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n \end{pmatrix}$ be row equivalent $m \times n$ matrices, where $\mathbf{a}_i$ and $\mathbf{b}_i$ are the i-th column of $\mathbf{A}$ and $\mathbf{B}$, respectively, for $i = 1, ..., n$. Show that for any $c_1, c_2, ..., c_n \in \mathbb{R}$,

   $$c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + \cdots + c_n\mathbf{a}_n = \mathbf{0}$$

   if and only if

   $$c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + \cdots + c_n\mathbf{b}_n = \mathbf{0}.$$

4. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ be a basis for a vector space V. Let $\mathbf{u}$ be a vector in V and let c be a scalar. Prove the following:

   (a) $[\mathbf{u} + \mathbf{v}]_S = [\mathbf{u}]_S + [\mathbf{v}]_S$.

   (b) $[c\mathbf{u}]_S = c\,[\mathbf{u}]_S$.

   (c) Suppose $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k$ are vectors in V. Note that for each $i = 1, 2, \ldots, k$, $[\mathbf{u}_i]_S$ is a vector in $\mathbb{R}^n$. By induction and using (a) and (b), it follows that if $c_1, c_2, \ldots, c_k \in \mathbb{R}$, then

   $$[c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \ldots + c_k\mathbf{u}_k]_S = c_1\,[\mathbf{u}_1]_S + c_2\,[\mathbf{u}_2]_S + \ldots + c_k\,[\mathbf{u}_k]_S\,.$$

   Prove that $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\}$ is linearly independent in V if and only if $\{[\mathbf{u}_1]_S\,, [\mathbf{u}_2]_S\,, \ldots, [\mathbf{u}_k]_S\}$ is linearly independent in $\mathbb{R}^n$.

NATIONAL UNIVERSITY OF SINGAPORE
Department of Mathematics

MA1522 Linear Algebra for Computing **Tutorial 7**

1. (a) Let $a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$ be a linear equation. Express this linear system as $\mathbf{a} \cdot \mathbf{x} = b$ for some (column) vectors $\mathbf{a}$ and $\mathbf{x}$.

   (b) Find the solution set of the linear system

$$\begin{array}{rcrcrcrcl} x_1 & + & 3x_2 & - & 2x_3 & & & = & 0 \\ 2x_1 & + & 6x_2 & - & 5x_3 & - & 2x_4 & = & 0 \\ & & & + & 5x_3 & + & 10x_4 & = & 0 \end{array}$$

   (c) Find a nonzero vector $\mathbf{v} \in \mathbb{R}^4$ such that $\mathbf{a}_1 \cdot \mathbf{v} = 0$, $\mathbf{a}_2 \cdot \mathbf{v} = 0$, and $\mathbf{a}_3 \cdot \mathbf{v} = 0$, where

$$\mathbf{a}_1 = \begin{pmatrix} 1 \\ 3 \\ -2 \\ 0 \end{pmatrix}, \qquad \mathbf{a}_2 = \begin{pmatrix} 2 \\ 6 \\ -5 \\ -2 \end{pmatrix}, \qquad \mathbf{a}_3 = \begin{pmatrix} 0 \\ 0 \\ 5 \\ 10 \end{pmatrix}.$$

   This exercise demonstrates the fact that if $\mathbf{A}$ is a $m \times n$ matrix, then the solution set of the homogeneous linear system $\mathbf{Ax} = \mathbf{0}$ consist of all the vectors in $\mathbb{R}^n$ that are orthogonal to every row vector of $\mathbf{A}$.

2. Let $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be an orthonormal set. Suppose

$$\mathbf{x} = \mathbf{v}_1 - 2\mathbf{v}_2 - 2\mathbf{v}_3 \quad \text{and} \quad \mathbf{y} = 2\mathbf{v}_1 - 3\mathbf{v}_2 + \mathbf{v}_3.$$

   Determine the value for each of the following

   (a) $\mathbf{x} \cdot \mathbf{y}$.

   (b) $||\mathbf{x}||$ and $||\mathbf{y}||$.

   (c) The angle θ between $\mathbf{x}$ and $\mathbf{y}$.

3. Let $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$, and $\mathbf{V} = \begin{pmatrix} \mathbf{v}_1 & \mathbf{v}_2 \end{pmatrix}$.

   (a) Compute $\mathbf{v}_1 \cdot \mathbf{v}_1$, $\mathbf{v}_1 \cdot \mathbf{v}_2$, $\mathbf{v}_2 \cdot \mathbf{v}_1$ and $\mathbf{v}_2 \cdot \mathbf{v}_2$.

   (b) Compute $\mathbf{V}^T\mathbf{V}$. What does the entries of $\mathbf{V}^T\mathbf{V}$ represent?

4. Let W be a subspace of $\mathbb{R}^n$. The *orthogonal complement* of W, denoted as $W^\perp$, is defined to be

$$W^\perp := \{\ \mathbf{v} \in \mathbb{R}^n \mid \mathbf{v} \cdot \mathbf{w} = 0 \text{ for all } \mathbf{w} \in W\ \}.$$

   Let $\mathbf{w}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$, $\mathbf{w}_2 = \begin{pmatrix} 1 \\ 2 \\ -1 \\ -2 \\ 0 \end{pmatrix}$, and $\mathbf{w}_3 = \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \\ 0 \end{pmatrix}$, and $W = \text{span}\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$.

(a) Show that $S = \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ is linearly independent.

(b) Show that S is orthogonal.

(c) Show that $W^\perp$ is a subspace of $\mathbb{R}^5$ by showing that it is a span of a set. What is the dimension? (**Hint**: See Question 1.)

(d) Obtain an orthonormal set T by normalizing $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$.

(e) Let $\mathbf{v} = \begin{pmatrix} 2 \\ 0 \\ 1 \\ 1 \\ -1 \end{pmatrix}$. Find the projection of $\mathbf{v}$ onto W.

(f) Let $\mathbf{v}_W$ be the projection of $\mathbf{v}$ onto W. Show that $\mathbf{v} - \mathbf{v}_W$ is in $W^\perp$.

This exercise demonstrated the fact that every vector $\mathbf{v}$ in $\mathbb{R}^5$ can be written as $\mathbf{v} = \mathbf{v}_W + \mathbf{v}_W^\perp$, for some $\mathbf{v}_W$ in W and $\mathbf{v}_W^\perp$ in $W^\perp$. In other words, $W + W^\perp = \mathbb{R}^5$.

5. Let $S = \{\mathbf{u_1}, \mathbf{u_2}, \mathbf{u_3}, \mathbf{u_4}\}$ where

$$\mathbf{u_1} = \begin{pmatrix} 1 \\ 2 \\ 2 \\ -1 \end{pmatrix}, \mathbf{u_2} = \begin{pmatrix} 1 \\ 1 \\ -1 \\ 1 \end{pmatrix}, \mathbf{u_3} = \begin{pmatrix} -1 \\ 1 \\ -1 \\ -1 \end{pmatrix}, \text{ and } \mathbf{u_4} = \begin{pmatrix} -2 \\ 1 \\ 1 \\ 2 \end{pmatrix}.$$

(a) Check that S is an orthogonal basis for $\mathbb{R}^4$.

(b) Is it possible to find a nonzero vector $\mathbf{w}$ in $\mathbb{R}^4$ such that $S \cup \{\mathbf{w}\}$ is an orthogonal set?

(c) Obtain an orthonormal set T by normalizing $\mathbf{u_1}, \mathbf{u_2}, \mathbf{u_3}, \mathbf{u_4}$.

(d) Let $\mathbf{v} = \begin{pmatrix} 0 \\ 1 \\ 2 \\ 3 \end{pmatrix}$. Find $[\mathbf{v}]_S$ and $[\mathbf{v}]_T$.

(e) Suppose $\mathbf{w}$ is a vector in $\mathbb{R}^4$ such that $[\mathbf{w}]_S = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 1 \end{pmatrix}$. Find $[\mathbf{w}]_T$.

Extra problems

1. Let $\mathbf{A}$ be an $m \times n$ matrix.

   (a) Show that the nullspace of $\mathbf{A}$ is equal to the nullspace of $\mathbf{A}^T\mathbf{A}$.

   (b) Show that $\text{nullity}(\mathbf{A}) = \text{nullity}(\mathbf{A}^T\mathbf{A})$ and $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^T\mathbf{A})$.

   (c) Is it true that $\text{nullity}(\mathbf{A}) = \text{nullity}(\mathbf{A}\mathbf{A}^T)$? Justify your answer.

   (d) Is it true that $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}\mathbf{A}^T)$? Justify your answer.

2. Let $\mathbf{A}$ and $\mathbf{B}$ be two matrices of the same size. Show that

$$\text{rank}(\mathbf{A} + \mathbf{B}) \leq \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{B}).$$

3. (a) Let W be a subspace of $\mathbb{R}^n$. Prove that the orthogonal complement of the orthogonal complement of W is W, i.e.

$$(W^\perp)^\perp = W.$$

   (b) Show that for any matrix $\mathbf{A}$, the column space of $\mathbf{A}$ is the orthogonal complement of the nullspace of $\mathbf{A}^T$,

$$\text{Col}(\mathbf{A})^\perp = \text{Null}(\mathbf{A}^T),$$

   or equivalently, the row space of $\mathbf{A}$ is the orthogonal complement of the nullspace of $\mathbf{A}$,

$$\text{Row}(\mathbf{A})^\perp = \text{Null}(\mathbf{A}).$$

NATIONAL UNIVERSITY OF SINGAPORE
Department of Mathematics

MA1522 Linear Algebra for Computing **Tutorial 8**

1. Apply Gram-Schmidt Process to convert

   (a) $\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 0 \\ 1 \end{pmatrix} \right\}$ into an orthonormal basis for $\mathbb{R}^4$.

   (b) $\left\{ \begin{pmatrix} 1 \\ 2 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 2 \\ 1 \end{pmatrix} \right\}$ into an orthonormal set. Is the set obtained an orthonormal basis? Why?

2. Let $\mathbf{A} = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & -1 & 1 & -1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 6 \\ 3 \\ -1 \\ 1 \end{pmatrix}$.

   (a) Is the linear system $\mathbf{Ax} = \mathbf{b}$ inconsistent?

   (b) Find a least squares solution to the system. Is the solution unique?

   (c) Use your answer in (b), compute the projection of $\mathbf{b}$ onto the column space of $\mathbf{A}$. Is the solution unique?

3. **(Application)** A line

$$p(x) = a_1 x + a_0$$

   is said to be the *least squares approximating line* for a given a set of data points (x_1, y_1), (x_2, y_2), ..., (x_m, y_m) if the sum

$$S = [y_1 - p(x_1)]^2 + [y_2 - p(x_2)]^2 + \cdots + [y_m - p(x_m)]^2$$

   is minimized. Writing

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix}, \ \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{pmatrix}, \text{ and } p(\mathbf{x}) = \begin{pmatrix} p(x_1) \\ p(x_2) \\ \vdots \\ p(x_m) \end{pmatrix} = \begin{pmatrix} a_1 x_1 + a_0 \\ a_1 x_2 + a_0 \\ \vdots \\ a_1 x_m + a_0 \end{pmatrix}$$

   the problem is now rephrased as finding a_0, a_1 such that

$$S = ||\mathbf{y} - p(\mathbf{x})||^2$$

is minimized. Observe that if we let

$$\mathbf{N} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_m \end{pmatrix} \text{ and } \mathbf{a} = \begin{pmatrix} a_0 \\ a_1 \end{pmatrix},$$

then $\mathbf{Na} = p(\mathbf{x})$. And so our aim is to find $\mathbf{a}$ that minimizes $||\mathbf{y} - \mathbf{Na}||^2$.

It is known the equation representing the dependency of the resistance of a cylindrically shaped conductor (a wire) at 20^oC is given by

$$R = \rho \frac{L}{A},$$

where R is the resistance measured in Ohms Ω, L is the length of the material in meters m, A is the cross-sectional area of the material in meter squared m^2, and ρ is the resistivity of the material in Ohm meters Ωm. A student wants to measure the resistivity of a certain material. Keeping the cross-sectional area constant at $0.002m^2$, he connected the power sources along the material at varies length and measured the resistance and obtained the following data.

L	0.01	0.012	0.015	0.02
R	2.75×10^{-4}	3.31×10^{-4}	3.92×10^{-4}	4.95×10^{-4}

It is known that the Ohm meter might not be calibrated. Taking that into account, the student wants to find a linear graph $R = \frac{\rho}{0.002} L + R_0$ from the data obtained to compute the resistivity of the material.

(a) Relabeling, we let $R = y$, $\frac{\rho}{0.002} = a_1$ and $R_0 = a_0$. Is it possible to find a graph $y = a_1 x + a_0$ satisfying the points?

(b) Find the least square approximating line for the data points and hence find the resistivity of the material. Would this material make a good wire?

4. **(Application)** Suppose the equation governing the relation between data pairs is not known. We may want to then find a polynomial

$$p(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$

of degree n, $n \leq m - 1$, that best approximates the data pairs (x_1, y_1), (x_2, y_2), ..., (x_m, y_m). A *least square approximating polynomial* of degree n is such that

$$||\mathbf{y} - p(\mathbf{x})||^2$$

is minimized. If we write

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix}, \ \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{pmatrix}, \ \mathbf{N} = \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^n \\ 1 & x_2 & x_2^2 & \cdots & x_2^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_m & x_m^2 & \cdots & x_m^n \end{pmatrix} \text{ and } \mathbf{a} = \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{pmatrix},$$

then $p(\mathbf{x}) = \mathbf{Na}$, and the task is to find $\mathbf{a}$ such that $||\mathbf{y} - \mathbf{Na}||^2$ is minimized. Observe that $\mathbf{N}$ is a matrix minor of the Vandermonde matrix. If at least $n+1$ of the x-values $x_1, x_2, ..., x_m$ are distinct, the columns of $\mathbf{N}$ are linearly independent, and thus $\mathbf{a}$ is uniquely determined by

$$\mathbf{a} = (\mathbf{N}^T\mathbf{N})^{-1}\mathbf{N}^T\mathbf{y}.$$

We shall now find a quartic polynomial

$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4$$

that is a least square approximating polynomial for the following data points

x	4	4.5	5	5.5	6	6.5	7	8	8.5
y	0.8651	0.4828	2.590	-4.389	-7.858	3.103	7.456	0.0965	4.326

Enter the data points.

```
>> x=[4 4.5 5 5.5 6 6.5 7 8 8.5]';
```

```
>> y=[0.8651 0.4828 2.590 -4.389 -7.858 3.103 7.456 0.0965 4.326]';
```

Next, we will generate the 10×10 Vandermonde matrix.

```
>> N=fliplr(vander(x));
```

We only want the matrix minor up to the 4-th power, that is, up to the the 5-th column,

```
>> N=N(:,1:5);
```

Use this to find the least square approximating polynomial of degree 4.

5. Let $\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$.

   (a) Find a QR factorization of $\mathbf{A}$.

   (b) Use your answer in (a) to find the least square solution to $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}$.

Extra problems

1. (a) Let S be the set of least square solutions to $\mathbf{Ax} = \mathbf{b}$. Show that there exists a $\mathbf{b}'$ such that $\mathbf{Av} = \mathbf{b}'$ for all $\mathbf{v} \in S$. This proves that the projection of $\mathbf{b}$ onto the column space of $\mathbf{A}$ is unique even though the least square solutions may not unique.

   (b) Suppose a linear system $\mathbf{Ax} = \mathbf{b}$ is consistent. Show that the solution set of $\mathbf{Ax} = \mathbf{b}$ is equal to the solution set of $\mathbf{A}^T\mathbf{Ax} = \mathbf{A}^T\mathbf{b}$.

2. (Uniqueness of orthogonal projection)
   Let V be a subspace of $\mathbb{R}^n$ and $\mathbf{u}$ a vector in $\mathbb{R}^n$. Show that $\mathbf{u}$ can be written uniquely as

   $$\mathbf{u} = \mathbf{u}_p + \mathbf{u}_n,$$

   such that $\mathbf{u}_n$ is a vector orthogonal to V and $\mathbf{u}_p$ is a vector in V. Remark: By the Gram-Schmidt process, $\mathbf{u}$ can be written as $\mathbf{u} = \mathbf{u}_p + \mathbf{u}_n$. We need to prove show that $\mathbf{u}_p$ and $\mathbf{u}_n$ are unique. Hint: if $\mathbf{u} = \mathbf{u}_p + \mathbf{u}_n = \mathbf{u}'_p + \mathbf{u}'_n$, where $\mathbf{u}_n, \mathbf{u}'_n$ are orthogonal to V and $\mathbf{u}_p, \mathbf{u}'_p \in V$, then $\mathbf{u}_n = \mathbf{u}'_n$ and $\mathbf{u}_p = \mathbf{u}'_p$).

3. Let $S = \{\mathbf{w}_1, ..., \mathbf{w}_k\}$ be an orthonormal basis for a subspace V in $\mathbb{R}^n$. Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in V.

   (a) Prove that $\mathbf{u} \cdot \mathbf{v} = [\mathbf{u}]_S \cdot [\mathbf{v}]_S$.

   (b) Prove that $\|[\mathbf{u}]_S\| = \|\mathbf{u}\|$.

   (c) Prove that the angle between $\mathbf{u}$ and $\mathbf{v}$ is equal to the angle between $[\mathbf{u}]_S$ and $[\mathbf{v}]_S$.

NATIONAL UNIVERSITY OF SINGAPORE
Department of Mathematics

MA1522 Linear Algebra for Computing **Tutorial 9**

1. A father wishes to distribute an amount of money among his three sons Jack, Jim, and John. He wish to distribute such that the following conditions are all satisfied.

   (i) The amount Jack receives plus twice the amount Jim receives is \$300.

   (ii) The amount Jim receives plus the amount John receives is \$300.

   (iii) Jack receives \$300 more than twice of what John receives.

   (a) Is it possible for the following conditions to all be satisfied?

   (b) If it is not possible, find a least square solution. (Make sure that your least square solution is feasible. For example, one cannot give a negative amount of money to anybody.)

2. (a) Suppose $\mathbf{A}$ is a $m \times n$ matrix where $m > n$. Let $\mathbf{A} = \mathbf{QR}$ be a QR factorization of $\mathbf{A}$. Explain how you might use this to write

   $$\mathbf{A} = \mathbf{Q}'\mathbf{R}',$$

   where $\mathbf{Q}'$ is an $m \times m$ orthogonal matrix, and $\mathbf{R}'$ a $m \times n$ matrix with $m - n$ zero rows at the bottom. This is known as the *full* QR factorization of $\mathbf{A}$.

   (b) In MATLAB, enter the following.

   ```
   >> A=sym([1 1 0;1 1 0;1 1 1;0 1 1])

   >> [Q R]=qr(A)
   ```

   What is $\mathbf{Q}$ and $\mathbf{R}$? Compare this with the answer in tutorial 8 question 5(a).

   (c) Explain how you might use the command `qr` in MATLAB to find a QR factorization of a $m \times n$ matrix $\mathbf{A}$?

3. (Cayley-Hamilton theorem)
   Consider

   $$p(\mathbf{X}) = \mathbf{X}^3 - 4\mathbf{X}^2 - \mathbf{X} + 4\mathbf{I}.$$

   (a) Compute $p(\mathbf{X})$ for $\mathbf{X} = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 1 \end{pmatrix}$.

   (b) Find the characteristic polynomial of $\mathbf{X}$.

   (c) Show that $\mathbf{X}$ invertible. Express the inverse of $\mathbf{X}$ as a function of $\mathbf{X}$.

This question demonstrated the *Cayley-Hamilton theorem*, which states that if $p(x)$ is the characteristic polynomial of $\mathbf{X}$, then $p(\mathbf{X}) = 0$. This also show that if 0 is not an eigenvalue of $\mathbf{X}$, then the constant term of the characteristic polynomial $p(x)$ is nonzero, and we can use that to compute the inverse of $\mathbf{X}$.

4. For each of the following matrices $\mathbf{A}$, determine if $\mathbf{A}$ is diagonalizable. If $\mathbf{A}$ is diagonalizable, find an invertible $\mathbf{P}$ that diagonalizes $\mathbf{A}$ and determine $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$.

   (a) $\mathbf{A} = \begin{pmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{pmatrix}$.

   (b) $\mathbf{A} = \begin{pmatrix} 9 & 8 & 6 & 3 \\ 0 & -1 & 3 & -4 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$

   (c) $\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$.

   (d) $\mathbf{A} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$

   (e) $\mathbf{A} = \begin{pmatrix} -1 & 1 & 1 \\ 1 & 1 & -1 \\ -4 & 2 & 3 \end{pmatrix}$.

5. (a) Show that λ is an eigenvalue of $\mathbf{A}$ if and only if it is an eigenvalue of $\mathbf{A}^T$.

   (b) Suppose $\mathbf{A}$ is diagonalizable. Is $\mathbf{A}^T$ diagonalizable? Justify your answer.

   (c) Suppose $\mathbf{v}$ is an eigenvector of $\mathbf{A}$ associated to eigenvalue λ. Show that $\mathbf{v}$ is an eigenvector of $\mathbf{A}^k$ associated to eigenvalue λ^k for any positive integer k.

   (d) If $\mathbf{A}$ is invertible, show that $\mathbf{v}$ is an eigenvector of $\mathbf{A}^k$ associated to eigenvalue λ^k for any negative integer k.

   (e) A square matrix is said to be *nilpotent* if there is a positive integer k such that $\mathbf{A}^k = \mathbf{0}$. Show that if $\mathbf{A}$ is nilpotent, then 0 is the only eigenvalue.

   (f) Let $\mathbf{A}$ be a $n \times n$ matrix with one eigenvalue λ with algebraic multiplicity n. Show that $\mathbf{A}$ is diagonalizable if and only if $\mathbf{A}$ is a scalar matrix, $\mathbf{A} = \lambda\mathbf{I}$.

   (g) Show that the only diagonalizable nilpotent matrix is the zero matrix.

Extra problems

1. Let $\mathbf{A}$ be an orthogonal matrix of order n and let $\mathbf{u}, \mathbf{v}$ be any two vectors in $\mathbb{R}^n$. Show that

   (a) $||\mathbf{u}|| = ||\mathbf{Au}||$;

   (b) $d(\mathbf{u}, \mathbf{v}) = d(\mathbf{Au}, \mathbf{Av})$; and

   (c) the angle between $\mathbf{u}$ and $\mathbf{v}$ is equal to the angle between $\mathbf{Au}$ and $\mathbf{Av}$.

2. Let $\mathbf{A}$ be an orthogonal matrix of order n. Let $S = \{\mathbf{u}_1, \mathbf{u}_2, ..., \mathbf{u}_k\}$ and define $T = \{\mathbf{Au_1}, \mathbf{Au_2}, ..., \mathbf{Au_k}\}$.

   (a) If S is orthogonal, show that T is orthogonal.

   (b) If S is orthonormal, is T orthonormal?

3. (a) Suppose $\mathbf{A}$ and $\mathbf{B}$ are similar matrices, that is, $\mathbf{A} = \mathbf{PBP}^{-1}$ for some invertible matrix $\mathbf{P}$. Then the characteristic polynomial of $\mathbf{A}$ and $\mathbf{B}$ are equal.

   (b) Suppose the characteristic polynomial of $\mathbf{A}$ and $\mathbf{B}$ are equal. Can we conclude that $\mathbf{A}$ and $\mathbf{B}$ are similar? Justify your answer.

   (c) Suppose $\mathbf{A}$ and $\mathbf{B}$ are $n \times n$ matrices with the same determinant. Is it true that their characteristic polynomials are equal? Justify your answer.

4. (a) Let $\mathbf{A}$ be a 2×2 matrix. Prove that the characteristic polynomial of $\mathbf{A}$ is

   $$x^2 - tr(\mathbf{A})x + \det(\mathbf{A}),$$

   where $tr(\mathbf{A})$ is the sum of the diagonal entries of $\mathbf{A}$.

   (b) Let $\mathbf{A}$ be a $n \times n$ matrix and $p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$. Prove that $a_0 = (-1)^n \det(\mathbf{A})$.

NATIONAL UNIVERSITY OF SINGAPORE
Department of Mathematics

MA1522 Linear Algebra for Computing **Tutorial 10**

1. A population of ants is put into a maze with 3 compartments labeled a, b, and c. If the ant is in compartment a, an hour later, there is a 20% chance it will go to compartment b, and a 40% change it will go to compartment c. If it is in compartment b, an hour later, there is a 10% chance it will go to compartment a, and a 30% chance it will go to compartment c. If it is in compartment c, an hour later, there is a 50% chance it will go to compartment a, and a 20% chance it will go to compartment b. Suppose 100 ants has been placed in compartment a.

   (a) Find the transition probability matrix $\mathbf{A}$. Show that it is a stochastic matrix.

   (b) By diagonalizing $\mathbf{A}$, find the number of ants in each compartment after 3 hours.

   (c) **(MATLAB)** We can use MATLAB to diagonalize the matrix $\mathbf{A}$. Type

   ```
   >> [P D]=eig(sym(A))
   ```

   The matrix $\mathbf{P}$ will be an invertible matrix, and $\mathbf{D}$ will be a diagonal matrix. Compare the answer with what you have obtained in (b).

   (d) In the long run (assuming no ants died), where will the majority of the ants be?

   (e) Suppose initially the numbers of ants in compartments a, b and c are α, β, and γ respectively. What is the population distribution in the long run (assuming no ants died)?

2. By diagonalizing $\mathbf{A} = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix}$, find a matrix $\mathbf{B}$ such that $\mathbf{B}^2 = \mathbf{A}$.

3. For each of the following symmetric matrices $\mathbf{A}$, find an orthogonal matrix $\mathbf{P}$ that orthogonally diagonalizes $\mathbf{A}$.

   (a) $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$.

   (b) $\mathbf{A} = \begin{pmatrix} 2 & 2 & -2 \\ 2 & -1 & 4 \\ -2 & 4 & -1 \end{pmatrix}$.

4. (**MATLAB**) Let $\mathbf{A} = \begin{pmatrix} 1 & -2 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & -2 & 1 \end{pmatrix}$.

   (a) Find an orthogonal matrix $\mathbf{P}$ that orthogonally diagonalizes $\mathbf{A}$, and compute $\mathbf{P}^T\mathbf{A}\mathbf{P}$.

(b) We will use MATLAB to orthogonally diagonalize $\mathbf{A}$. Type

```
>> A=[1 -2 0 0;-2 1 0 0;0 0 1 -2;0 0 -2 1];

>> [P D]=eig(A);

>> sym(P), sym(D)
```

Compare the result with your answer in (a).

5. Find the SVD of the following matrices $\mathbf{A}$.

(a) $\mathbf{A} = \begin{pmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{pmatrix}$.

(b) $\mathbf{A} = \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix}$.

(c) $\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}$.

6. (**MATLAB**) Let $\mathbf{A} = \begin{pmatrix} -18 & 13 & -4 & 4 \\ 2 & 19 & -4 & 12 \\ -14 & 11 & -12 & 8 \\ -2 & 21 & 4 & 8 \end{pmatrix}$.

(a) Find a SVD of $\mathbf{A}$.

(b) In MATLAB, type

```
>> [U S V]=svd(A)
```

Compare the result with your answer in (a).

Extra problems

1. Let $\mathbf{A}$ be a stochastic matrix. Prove that $\lambda = 1$ is an eigenvalue of $\mathbf{A}$.

2. Let $\mathbf{v}_1$ be an eigenvector of $\mathbf{A}$ associated to the eigenvalue λ_1 and $\mathbf{v}_2$ an eigenvector of $\mathbf{A}^T$ associated to eigenvalue λ_2. Suppose $\lambda_1 \neq \lambda_2$. Show that v_1 and v_2 are orthogonal.

3. Let $\mathbf{A}$ be an $n \times n$ matrix. Show that there exists an orthogonal matrix $\mathbf{Q}$ such that

$$\mathbf{A}\mathbf{A}^T = \mathbf{Q}^T\mathbf{A}^T\mathbf{A}\mathbf{Q}$$

NATIONAL UNIVERSITY OF SINGAPORE
Department of Mathematics

MA1522 Linear Algebra for Computing **Tutorial 11**

1. (i) Determine whether the following are linear transformations.
   (ii) Write down the standard matrix for each other the linear transformations.
   (iii) Find a basis for the range for each of the linear transformations.
   (iv) Find a basis for the kernel for each of the linear transformations.

   (a) $T_1\colon \mathbb{R}^2 \to \mathbb{R}^2$ such that $T_1\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x+y \\ y-x \end{pmatrix}$ for $\begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2$.

   (b) $T_2\colon \mathbb{R}^2 \to \mathbb{R}^2$ such that $T_1\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} 2^x \\ 0 \end{pmatrix}$ for $\begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2$.

   (c) $T_3\colon \mathbb{R}^2 \to \mathbb{R}^3$ such that $T_3\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x+y \\ 0 \\ 0 \end{pmatrix}$ for $\begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2$.

   (d) $T_4\colon \mathbb{R}^3 \to \mathbb{R}^3$ such that $T_4\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = \begin{pmatrix} 1 \\ y-x \\ y-z \end{pmatrix}$ for $\begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3$.

   (e) $T_5\colon \mathbb{R}^5 \to \mathbb{R}$ such that $T_5\left(\begin{pmatrix} x_1 \\ \vdots \\ x_5 \end{pmatrix}\right) = x_3 + 2x_4 - x_5$ for $\begin{pmatrix} x_1 \\ \vdots \\ x_5 \end{pmatrix} \in \mathbb{R}^5$.

   (f) $T_6 : \mathbb{R}^n \to \mathbb{R}$ such that $T_6(\mathbf{x}) = \mathbf{x} \cdot \mathbf{x}$ for $\mathbf{x} \in \mathbb{R}^n$.

2. Let $F : \mathbb{R}^3 \to \mathbb{R}^3$ and $G : \mathbb{R}^3 \to \mathbb{R}^3$ be linear transformations such that

$$F\left(\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}\right) = \begin{pmatrix} x_1 - 2x_2 \\ x_1 + x_2 - 3x_3 \\ 5x_2 - x_3 \end{pmatrix} \text{ and } G\left(\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}\right) = \begin{pmatrix} x_3 - x_1 \\ x_2 + 5x_1 \\ x_1 + x_2 + x_3 \end{pmatrix},$$

   and let $\mathbf{A}_F$ and $\mathbf{B}_G$ be the standard matrix of F and G, respectively.

   (a) Find $\mathbf{A}_F$ and $\mathbf{B}_G$.

   (b) Define

$$(F+G)(\mathbf{x}) := F(\mathbf{x}) + G(\mathbf{x}), \text{ for all } \mathbf{x} \in \mathbb{R}^3.$$

   Is $(F+G)$ a linear transformation? If it is, find its standard matrix.

   (c) Write down the formula for $F(G(\mathbf{x}))$ and find its standard matrix.

   (d) Find a linear transformation $H : \mathbb{R}^3 \to \mathbb{R}^3$ such that

$$H(G(\mathbf{x})) = \mathbf{x}, \quad \text{for all } \mathbf{x} \in \mathbb{R}^3.$$

3. For each of the following linear transformations, (i) determine whether there is enough information for us to find the formula of T; and (ii) find the formula and the standard matrix for T if possible.

   (a) $T : \mathbb{R}^3 \to \mathbb{R}^4$ such that

$$T\left(\begin{pmatrix}1\\0\\0\end{pmatrix}\right) = \begin{pmatrix}1\\3\\0\\1\end{pmatrix}, \; T\left(\begin{pmatrix}0\\1\\0\end{pmatrix}\right) = \begin{pmatrix}2\\2\\-1\\4\end{pmatrix}, \text{ and } T\left(\begin{pmatrix}0\\0\\1\end{pmatrix}\right) = \begin{pmatrix}0\\4\\1\\6\end{pmatrix}.$$

   (b) $T : \mathbb{R}^2 \to \mathbb{R}^2$ such that

$$T\left(\begin{pmatrix}1\\-1\end{pmatrix}\right) = \begin{pmatrix}2\\0\end{pmatrix}, \; T\left(\begin{pmatrix}1\\1\end{pmatrix}\right) = \begin{pmatrix}0\\2\end{pmatrix}, \text{ and } T\left(\begin{pmatrix}2\\0\end{pmatrix}\right) = \begin{pmatrix}2\\2\end{pmatrix}.$$

   (c) $T : \mathbb{R}^3 \to \mathbb{R}$ such that

$$T\left(\begin{pmatrix}1\\-1\\0\end{pmatrix}\right) = -1, \; T\left(\begin{pmatrix}0\\1\\-1\end{pmatrix}\right) = 1 \text{ and } T\left(\begin{pmatrix}-1\\0\\1\end{pmatrix}\right) = 0.$$

4. For each of the following linear transformations T, determine its rank and nullity, and whether it is one-to-one, and/or onto.

   (a) T: $\mathbb{R}^4 \to \mathbb{R}^6$ such that the rank is 4.

   (b) T: $\mathbb{R}^6 \to \mathbb{R}^4$ such that the nullity is 2.

   (c) $T : \mathbb{R}^4 \to \mathbb{R}^6$ such that the reduce row-echelon form of its standard matrix has 3 nonzero rows.

   (d) $T : \mathbb{R}^3 \to \mathbb{R}^3$ such that T is one-to-one.

Extra problems

1. (a) Show that if λ is an eigenvalue of a stochastic matrix $\mathbf{P}$, then $|\lambda| \leq 1$.

   Hint: Pick an eigenvector $\mathbf{v}$ of $\mathbf{P}^T$ associated with λ. Let $k \in \{1, 2, \ldots, n\}$ be a coordinate of $\mathbf{v}$ with the maximum absolute value, $|v_k| \geq |v_i|$ for all $i = 1, \ldots, n$. Consider the k-th coordinate of the equation $\mathbf{P}^T\mathbf{v} = \lambda\mathbf{v}^T$.

   (b) Let $\mathbf{P}$ be a stochastic matrix. For any vector $\mathbf{v}$, define $\mathbf{v}^{(k)} = \mathbf{P}^k\mathbf{v}$. Show that if $\mathbf{v}$ is an eigenvector of $\mathbf{P}$ that is not associated to eigenvalue 1, then $\mathbf{v}^{(k)} \to 0$ as $k \to \infty$.

2. Let $\{\mathbf{u}_1, ..., \mathbf{u}_n\}$ be a basis for $\mathbb{R}^n$ and $\{\mathbf{v}_1, ..., \mathbf{v}_n\}$ be some vectors in $\mathbb{R}^m$. Prove that there is a unique transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ such that $T(\mathbf{u}_i) = \mathbf{v}_i$ for $i = 1, ..., n$.

3. Prove that $T : \mathbb{R}^n \to \mathbb{R}^n$ is a bijective (one-to-one and onto) linear transformation if and only if there is a basis $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ such that the standard matrix of T is a transition matrix from S to E, where E is the standard basis for $\mathbb{R}^n$.